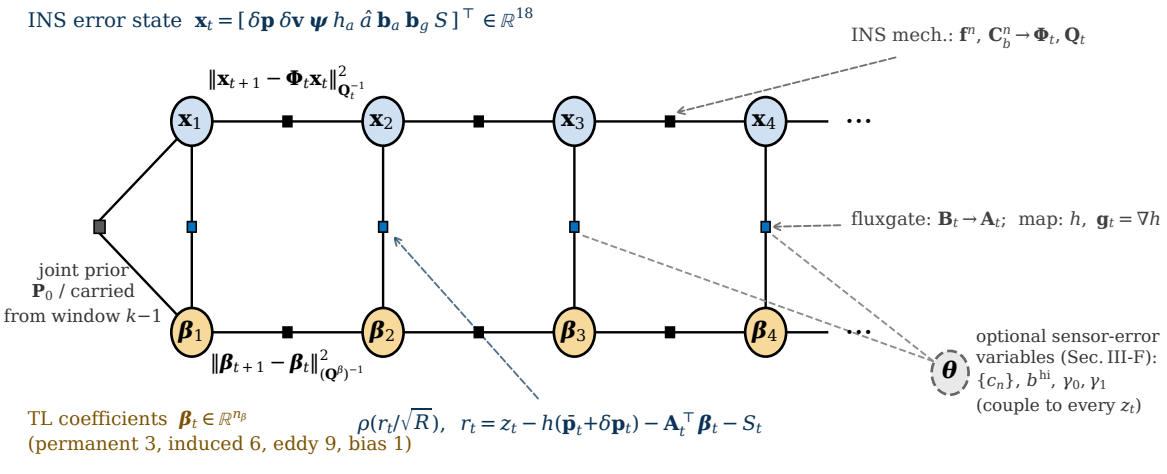
INS error state $\mathbf{x}_t = [\,\delta\mathbf{p}\ \delta\mathbf{v}\ \boldsymbol{\psi}\ h_a\ \hat{a}\ \mathbf{b}_a\ \mathbf{b}_g\ S\,]^\top \in \mathbb{R}^{18}$
INS mech.: $\mathbf{f}^n,\ \mathbf{C}_b^n \to \boldsymbol{\Phi}_t,\ \mathbf{Q}_t$
$\|\mathbf{x}_{t+1} - \boldsymbol{\Phi}_t\mathbf{x}_t\|^2_{\mathbf{Q}_t^{-1}}$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$
fluxgate: $\mathbf{B}_t \to \mathbf{A}_t$; map: $h$, $\mathbf{g}_t = \nabla h$
joint prior $\mathbf{P}_0$ / carried from window $k-1$
$\boldsymbol{\beta}_1$
$\boldsymbol{\beta}_2$
$\boldsymbol{\beta}_3$
$\boldsymbol{\beta}_4$
$\|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}}$
$\boldsymbol{\theta}$
optional sensor-error variables (Sec. III-F): $\{c_n\}$, $b^{\mathrm{hi}}$, $\gamma_0$, $\gamma_1$ (couple to every $z_t$)
TL coefficients $\boldsymbol{\beta}_t \in \mathbb{R}^{n_\beta}$
(permanent 3, induced 6, eddy 9, bias 1)
$\rho(r_t/\sqrt{R}),\quad r_t = z_t - h(\tilde{\mathbf{p}}_t + \delta\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t$